Saif Khan

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TECHNICAL SKILLS

Python, C++, SQL, Git, HTML, CSS, MLOps, Docker, AWS, Apache Airflow

EXPERIENCE

Data Science Intern · **Suprmentr Technologies Pvt.Ltd.** Feb-May 2026

Engineering Employability Analytics: What Drives Graduate Success in India? | Python· Pandas· XGBoost· SHAP· Streamlit

• Designed and developed an end-to-end employability analytics platform on **56,000+** engineering student records to investigate key drivers of graduate placement and salary outcomes.
• Engineered and evaluated an **XGBoost** prediction model (ROC-AUC **0.82**, 5-fold CV 0.81 ± 0.02), using **SHAP** explainability to reveal that internships and college tier influence employability more strongly than CGPA.
• Delivered actionable insights including a **42%** salary premium for Computer Science graduates and deployed a public-facing Streamlit dashboard for interactive career trend exploration.

PROJECTS

Where Does India's ₹33 Lakh Crore Tax Revenue Go? | Python · Scikit-learn · Streamlit · NLP

• Led the development of a public-finance analytics platform by consolidating **18 years** of government taxation, expenditure, and state finance data into a unified analytical framework.
• Engineered fiscal performance indicators and applied K-Means clustering, Holt-Winters forecasting, and statistical analysis to identify revenue concentration, taxpayer participation trends, and state-level development patterns.
• Uncovered key insights including **60%** GST concentration across **six states** and taxpayer participation below **2%** of India's population, highlighting structural imbalances within the tax ecosystem.
• Designed and deployed an interactive Streamlit dashboard enabling exploration of **India's ₹33 lakh crore tax flow**, ministry spending efficiency, and long-term fiscal trends through data-driven visualizations.

How Global Oil Crises Impact India's Economy | Python · Pandas · Time-series · nltk · arima · Streamlit

• Built an end-to-end economic intelligence platform using **six years** of geopolitical, crude oil, exchange rate, and trade datasets to analyze how global conflicts influence India's economic performance.
• Applied time-series forecasting, statistical modeling, and correlation analysis to quantify economic risks, finding that every **$10** increase in crude oil prices **weakens** the **Indian Rupee** and increases annual import costs by approximately **$15 billion**.
• Developed an ARIMA forecasting model (MAE: $0.25/barrel) and deployed an interactive Streamlit application enabling exploration of oil-price scenarios, inflation risks, and trade-cost projections.

Crowd Safety & Pilgrim Forecasting for Hajj Operations | Python· Pandas· Scikit-learn· Streamlit

• Designed and developed a predictive analytics platform leveraging **26 years** of Hajj and Umrah data to evaluate crowd safety risks and forecast future pilgrimage demand.
• Engineered features from pilgrim volume, incident, and environmental datasets, applying statistical analysis and machine learning techniques to identify crowd density and extreme heat as primary contributors to **safety-related events**.
• Built and compared Machine Learning and forecasting models including Random Forest, Gradient Boosting, and ARIMA; achieved best performance with Random Forest ($R^2 = 0.75$) and forecasted **~1.9 Millon pilgrims** for 2026.
• Delivered insights through a deployed Streamlit dashboard that allows users to explore historical patterns, risk indicators, and demand forecasts for **large-scale pilgrimage event**s.

EDUCATION

B.E. — Computer Science & Engineering, **Navodaya Institute of Technology** May 2026

ACHIEVEMENTS & PORTFOLIO

• Built and deployed 4 public-facing Data Science applications using Machine Learning, Streamlit, and Python, transforming complex datasets into interactive decision-support tools.
• Portfolio showcases live analytics platforms covering Indian education outcomes, public finance, economic risk analysis, and crowd safety forecasting.
• Published complete project source code, dashboards, and documentation through GitHub and personal portfolio website.